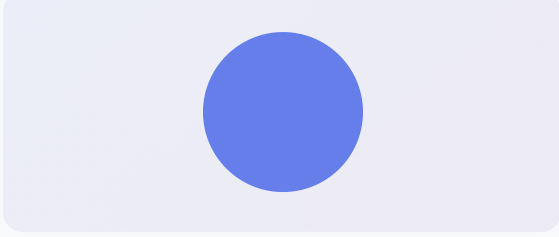


CSS Border-Radius

Project Examples & Implementation Guide

1. Basic Shape Examples

Perfect Circle

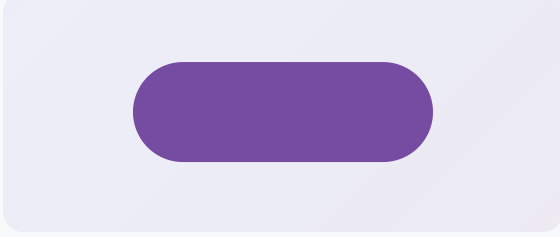


```
border-radius: 50%;
```

Used for profile pictures, avatars, and badge indicators.

Use Case: User avatars, circular buttons, loading spinners

Pill/Capsule



```
width: 150px; height: 50px; border-radius: 25px;
```

Creates a rounded rectangle, perfect for buttons and tags.

Use Case: Call-to-action buttons, filter tags, badges

Rounded Square



```
border-radius: 15px;
```

Slightly rounded corners for a modern, polished look.

Use Case: Card components, image frames, containers

2. Advanced Shape Techniques

Variable Corners



```
border-radius: 45px 10px 45px 10px;
```

Different radius values for each corner create unique effects.

Use Case: Design elements, custom decorative shapes

Organic/Blob Shape

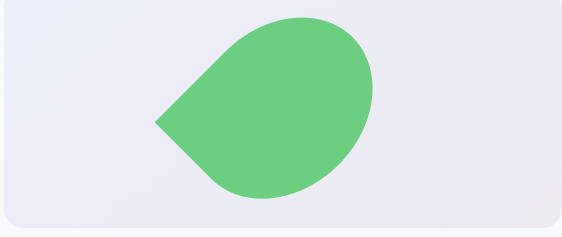


```
border-radius: 60% 40% 30% 70% / 60% 30% 70% 40%;
```

Creates asymmetrical, blob-like shapes for modern designs.

Use Case: Hero sections, decorative elements, SVG alternatives

Teardrop Shape



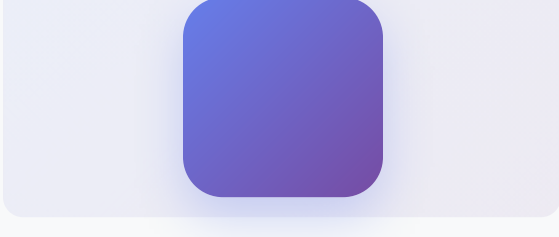
```
border-radius: 50% 50% 50% 0%; transform: rotate(45deg);
```

Combines border-radius with transform for creative shapes.

Use Case: Location pins, pointers, creative illustrations

3. Real-World Project Applications

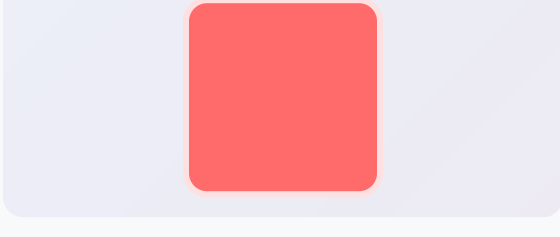
Mobile App UI Cards



```
border-radius: 20px; box-shadow: 0 10px 25px rgba(...);
```

Material Design cards with elevation and rounded corners.

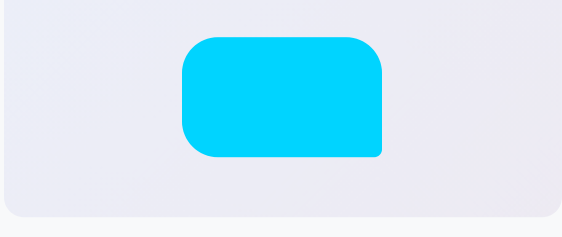
Product Image Gallery



```
border-radius: 12px; border: 3px solid;
```

Images with subtle borders and rounded corners for e-commerce.

Chat Bubbles



```
border-radius: 18px 18px 4px 18px;
```

Messaging apps use asymmetric corners for chat bubbles.

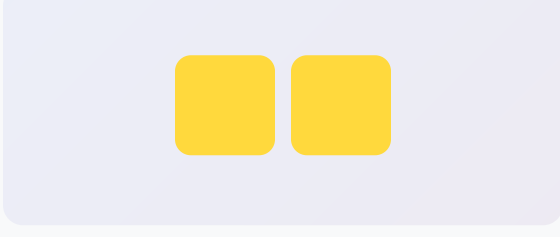
Toggle Switches



```
border-radius: 25px; /* pill */ /* applied to icon container */ .toggle: 50%; /* circle */
```

Combination of pill shape and circles for UI controls.

Rating Stars



```
border-radius: 8px; /* applied to icon container */
```

Rounded containers holding star ratings and reviews.

Color Palette Picker



```
/* Varying radius values */ border-radius: 50% / 20% / 0%;
```

Interactive color swatches with different border radius values.

4. Advanced CSS Techniques

Pro Tips & Best Practices

Symmetrical vs Asymmetrical: Use 4 values for different corners, 8 values for horizontal/vertical control

Slash Syntax: The "/" separator allows independent control of horizontal and vertical radii (e.g., 50% / 30%)

Percentage vs Pixels: Use % for responsive designs, px for fixed dimensions

Overflow Hidden: Combine with overflow:hidden to properly clip content inside rounded corners

Transition Smoothness: Add transition: border-radius 0.3s for smooth animations

Performance: border-radius is GPU-accelerated; safe to use with animations

Combining with Shadows: Round corners + box-shadow creates depth and modern aesthetics

Border Consideration: Borders respect border-radius; use border + border-radius for outlined shapes

CSS Syntax Reference

```
/* Four values - top-left, top-right, bottom-right, bottom-left */ border-radius: 10px 20px 30px 40px; /* Horizontal / Vertical */ border-radius: 50% / 30%; /* Individual properties */ border-top-left-radius: 20px; border-top-right-radius: 30px; border-bottom-right-radius: 20px; border-bottom-left-radius: 10px; /* Animation */ border-radius: 20px; transition: border-radius 0.3s ease; /* On hover effect */ &:hover { border-radius: 50%; }
```

5. Common Mistakes to Avoid

✗ Mistake: Overflow Issues

```
border-radius: 20px; /* Content overflows! */
```

Fix: Add overflow: hidden; to clip content properly inside rounded corners.

✗ Mistake: Forgetting Box-Sizing

```
width: 100px; height: 100px; border: 5px solid; border-radius: 50%; /* not perfectly round */
```

Fix: Use box-sizing: border-box; for consistent calculations.

⚠ Performance: Too Many Borders

```
border-radius: 60% 40% 30% 70% / 60% 30% 70% 40%; border: complex gradient;
```

Tip: Complex shapes with gradients can impact performance. Test on low-end devices.

⚠ Browser Compatibility

```
/* Old syntax (deprecated) */ border-radius: 50%; /* Works in all modern browsers */
```

Note: border-radius is well-supported. Avoid vendor prefixes for new projects.

6. Browser Support & Compatibility

Chrome: ✓ Full Support (v4+) Firefox: ✓ Full Support (v4+) Safari: ✓ Full Support (v5+) Edge: ✓ Full Support (All versions) IE 11: ✓ Basic Support (no gradients in corners) Mobile: ✓ Full Support (iOS Safari, Android Chrome)

Note: border-radius is now a standard CSS property with excellent cross-browser support. No vendor prefixes needed for modern projects. For legacy browser support (IE 8 and below), consider using fallback images or SVG shapes.